

Structural Admissibility and Biological Viability Uniquely Select the Standard Genetic Code

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2026

Preprint.

Archival version (Zenodo): [DOI pending publication]

Significance

We show that the standard genetic code is the unique survivor of a two-stage sieve over the space of all wobble-admissible codon tables, under eight simultaneous biological viability criteria. Stage 1 (wobble admissibility) reduces the search space by 10^{50} . Stage 2 applies five classical criteria simultaneously — placing the standard code at $z = +3.22\sigma$, top 0.009% of 10^5 random codes — and three novel criteria (evolvability, historical reachability, complete stop coverage). No code in 10^5 samples outperforms the standard code under all eight criteria. A CP-SAT global search reveals that codes with higher abstract scores are biologically disqualified: they have hyper-conservative mutation profiles that would prevent drastic substitutions required for protein fold diversification. This is the first computational uniqueness result for the genetic code across more than five simultaneous criteria.

Status of Claims

Note: the categories below are local to this paper and apply to the biological analysis. For the three GTE arithmetic echoes (§4.2), the standard UGP taxonomy applies: those are Lean-certified [T] exact arithmetic results.

Category A: Derived analytically.

Wobble admissibility criterion; Stage-1 reduction by 10^{50} ; mathematical formulation of Stages 2A–2I; the observation that CP-SAT optimal codes have $\max_jump \leq 5$; all z-score computations.

Category B (bridge): Computationally established, sampling-based.

The uniqueness result (0/ 10^5 codes beat standard under all eight criteria); z-score $+3.22\sigma$; all phase-by-phase z-score values; variant-code comparisons. These are sampling results, not analytic proofs.

Category C: Pipeline choices.

Equal weights for the five classical criteria; the

threshold 0.56 for Stage 2H; specific prebiotic abundance rankings for first-wave amino acids; the value $\max_jump_{\min} = 6.0$ for Stage 2G.

Category D: Open/speculative.

The claim that uniqueness holds for *all* 5.8×10^{36} complete wobble-admissible codes (not just the 10^5 sampled); the connection to MDL over prebiotic reaction networks; the formal algebraic proof via Tarski-Seidenberg.

Abstract

The near-universality of the standard genetic code raises a deep question: is it the best possible encoding of amino acids, or merely one of many adequate solutions? We address this by applying a two-stage sieve — admissibility followed by multi-criterion viability — to the space of all codon-to-amino-acid maps. **Stage 1** (wobble admissibility) reduces the 23^{64} possible maps by a factor of 10^{50} , to the $23^{27} \approx 5.8 \times 10^{36}$ maps that respect the Crick wobble degeneracy rules.

Stage 2 applies eight independent biological viability criteria simultaneously: error minimization, prebiotic accessibility, stop-codon robustness, chemical clustering, polar-requirement conservation, evolvability (minimum accessible mutation diversity), historical reachability (first-wave amino acids in ancient codon boxes), and complete stop-codon coverage. Monte Carlo sampling of 10^5 complete wobble-admissible codes shows that the standard code ranks at $z = +3.22\sigma$ on the five-criterion Phase 5 metric ($p < 0.009\%$). Global CP-SAT optimization confirms the standard code is not the mathematical maximum of any single or combined abstract criterion. However, codes that score higher on abstract criteria have $\max_jump \leq 5$ (hyper-conservative mutation profiles), which disqualifies them under the evolvability criterion (Stage 2G).

With all eight criteria applied jointly, no code among 10^5 randomly generated complete wobble-admissible codes outperforms the standard code. The standard genetic code is the **unique survivor** of the two-stage sieve within this sampling depth. Codes that score higher on abstract criteria all carry hyper-conservative mutation profiles ($\max_jump \leq 5$), which disqualifies them

on biological grounds: they would prevent the drastic substitutions required for protein fold diversification.

1 Introduction

The genetic code maps 64 codons to 20 amino acids and 3 stop signals. Its near-universality across all domains of life is remarkable and has long been taken as evidence that the code is not accidental. Yet explaining why *this particular* code, rather than any of the $23^{64} \approx 10^{85}$ conceivable alternatives, was selected remains an open question.

Existing explanations fall into three categories. The *stereochemical hypothesis* [25, 13] proposes direct chemical affinity between codons/anticodons and amino acids. The *co-evolution hypothesis* [26] links codon assignments to the biosynthetic relationships among amino acids. The *error-minimization hypothesis* [3, 5] shows that the code minimizes the physicochemical distance of amino acids encoded by synonymous codons, thereby buffering translation errors.

The most influential quantitative result is that of Freeland and Hurst [3], who showed that the standard code outperforms approximately 99% of random codes on a single error-minimization criterion. Subsequent work [8, 4, 9] extended this analysis. Recent evolutionary-dynamics work argues that “near-ideal selection” operates during code escape and diaspora phases, efficiently selecting codes supporting rapid cellular division [27], consistent with the near-optimality demonstrated here. No prior study has simultaneously applied more than a small number of criteria or performed a global search over the full code space.

We apply a two-stage sieve framework to genetic code selection. Stage 1 (structural admissibility) corresponds to wobble decodability — a combinatorial quotient structure on codon space. Stage 2 (viability) applies multiple biochemical fitness criteria simultaneously. This framework is motivated by a pattern identified in companion work on the Standard Model [21, 22]: the intersection of a structural admissibility condition with a viability condition can uniquely select a single configuration from an astronomically large candidate space. The present paper tests whether this pattern extends to biology.

Section 4.2 and Appendix B document three arithmetic echoes between the UGP/GTE canonical orbit at $n = 10$ and fundamental counts in the genetic code: the Mersenne ratio $c_3/c_2 \approx 64$ (codons), the prime-lock residue $m_1 = 20$ (amino acids), and $a_2 = 3^2$ with wobble classes = 3^3 . All three are Lean 4-certified exact arithmetic results [24], not model fits, and are independent of the main sieve analysis.

Our contributions are:

1. A wobble-admissibility framework that reduces the search space by 10^{50} .
2. A five-criterion joint viability analysis placing the standard code at $z = +3.22\sigma$ (top 0.009%).
3. CP-SAT global optimization characterizing codes that score higher on abstract metrics and showing why they are biologically disqualified (hyper-conservative mutation profiles, Stage 2G).
4. Three additional viability criteria (evolvability, historical reachability, complete stop coverage) that reduce competitors from 9 to 0 in a sample of 10^5 complete codes.
5. A sampling-based uniqueness result: 0/99,998 evolvable complete codes outperform the standard under all eight criteria simultaneously.

2 A Two-Stage Admissibility-Viability Sieve for the Genetic Code

2.1 Stage 1: Wobble Admissibility

The 64 codons (NNN , $N \in \{U, C, A, G\}$) form a natural equivalence structure under the Crick wobble rules [1]: the third-position base of a codon is partially degenerate, collapsing the 64-codon space into 27 *wobble equivalence classes*. A codon-to-amino-acid map is **wobble admissible** if and only if all codons within the same wobble class encode the same amino acid (or stop signal).

Wobble admissibility reduces the code space from $23^{64} \approx 10^{85}$ total maps to $23^{27} \approx 5.8 \times 10^{36}$, a reduction by a factor of $\sim 10^{50}$. All subsequent analysis is restricted to this admissibility-reduced space.

2.2 Stage 2: Eight Viability Criteria

Within the admissible space, we apply eight independent criteria. Each criterion operationalizes a distinct biochemical argument for why certain codon assignments are more viable than others.

Stage 2A: Error minimization [3]. The average polar-requirement (PR) difference [25] between amino acids encoded by codons differing in one nucleotide (all Hamming-1 pairs). Lower values indicate that translation errors have smaller physicochemical consequences. The standard code has error score = 1.684.

Stage 2B: Prebiotic accessibility. A code is more viable if the codon families with more members encode the amino acids that were most abundant in prebiotic chemistry [11, 2]. The prebiotic accessibility score rewards codes in which the most prebiotically common amino acids (Gly, Ala, Asp, Glu, Val, Ser) occupy the largest codon families [7].

Stage 2C: Stop-codon robustness. The code must provide ≥ 2 distinct stop-codon positions. This ensures robust translation termination and prevents readthrough at a single stop site. All standard variant codes (mitochondrial, ciliate, etc.) are checked against this criterion.

Stage 2D: Chemical clustering. Amino acids encoded by Hamming-1 neighbours should be chemically similar (PR difference < 3 units). This operationalizes the error robustness principle at the single-step level. The standard code achieves a clustering score of 0.714 (fraction of Hamming-1 pairs within 3 PR units).

Stage 2F: Polar-requirement jump conservation. The maximum PR difference across any Hamming-1 sense-sense pair (*max_jump*) must not exceed the standard code’s value of 7.40. This prevents biologically catastrophic single-mutation substitutions while allowing the observed range of amino acid diversity [28].

Stage 2G: Evolvability (minimum accessible mutation diversity). A viable code must also permit a *minimum* level of amino acid diversity accessible via single-nucleotide mutation: *max_jump* ≥ 6.0 . Codes with *max_jump* < 6 prevent all substitutions that change PR by more than 5 units. This class includes evolutionarily critical mutations including the Glu→Val substitution underlying sickle-cell haemoglobin [6] ($\Delta\text{PR} = 6.9$) and the Asp→Asn class found in adaptive substitutions in many kinases [10]. A code that forbids all such drastic substitutions would be too constrained for the evolution of complex enzymatic functions and protein folds.

Stage 2H: Historical reachability. Under the co-evolution hypothesis [26], the code was assembled incrementally starting from the most prebiotically accessible amino acids. Operationally: the fraction of the seven most prebiotically abundant amino acids (Gly, Ala, Asp, Glu, Val, Ser, Pro) that occupy four-fold degenerate codon boxes (the most redundant and evolutionarily ancient assignments) must be ≥ 0.56 . The standard code achieves 0.714 on this criterion.

Stage 2I: Complete stop-codon coverage. All three stop-codon identities (UAA, UAG, UGA) must appear at least once in the code. Codes with only two stop signals lack the full complement recognized by the two release factors RF1 (UAA, UAG) and RF2 (UAA, UGA) [16]. Incomplete stop coverage is evolutionarily unstable: the absent stop codon can be captured by a near-cognate tRNA and reassigned to an amino acid, reducing termination reliability over evolutionary time.

3 Results

3.1 Monte Carlo analysis (Phases 1–5)

We generated 10^5 random wobble-admissible complete codes using the rejection-free construction of [19], ensuring all 20 sense amino acids appear and at least 2 stop signals are present. For each code we computed the five-criterion score (Stages 2A–2D, 2F):

$$\text{Score}_5 = -w_1 \cdot \text{err} + w_2 \cdot \text{acc} + w_3 \cdot \text{clus} - w_4 \cdot \frac{\text{max_jump}}{10}$$

with equal weights $w_i = 1$.

The standard code scores -1.257 on this metric. Among 10^5 random complete codes, only 9 beat the standard (0.009%), placing it at $z = +3.22\sigma$ above the mean. This extends the Freeland–Hurst result [3] from one criterion ($z \approx -2.0\sigma$, top $\sim 1\%$) to five simultaneous criteria (top 0.009%).

3.2 Phase-by-phase accumulation of z-score

Table 1 shows how the z-score builds as each criterion is added.

Table 1: Standard code z-score by phase of analysis.

Phase	Criteria	z-score
1: Error (Freeland-Hurst)	2A	-2.04σ
2: +accessibility	2A, 2B	$+2.22\sigma$
3: +stop, +clustering	2A–2D	$+2.43\sigma$
4: +completeness filter	2A–2D	$+3.59\sigma$
5: +max-jump	2A–2D, 2F	$+3.76\sigma$
6: revised metric	2A–2D, 2F	$+3.22\sigma$

3.3 CP-SAT global search

We applied the Google OR-Tools CP-SAT solver to find the globally optimal code for each objective function. For the 4-criterion objective with *max_jump* as a hard constraint, CP-SAT found codes scoring -0.051 (vs. standard code -0.338). These codes uniformly have *max_jump* ≈ 4 – 5 — hyper-conservative codes where no single-nucleotide mutation can change the polar requirement by more than 5 units.

This confirms that the standard code is *not* the unique mathematical maximum of any abstract combination of criteria we tested. However, the CP-SAT codes violate Stage 2G (evolvability): all have *max_jump* < 6.0 .

3.4 Variant code confirmations

All known variant genetic codes (vertebrate and invertebrate mitochondrial, yeast mitochondrial, ciliate, Mycoplasma, etc.) were tested. All score *below* the standard code on the joint Phase-5 metric and all score below all

local optima of the metric. The mitochondrial codes (which reassign one or two codons) are confirmed as local minima of the metric: each variant is reached from the standard code by a single codon reassignment that decreases the score.

3.5 Phase 6: Uniqueness under eight criteria

The full numerical breakdown appears in Tables 2 and 3. Adding Stages 2G, 2H, and 2I sequentially:

- **+Stage 2G (evolvability, $\text{max_jump} \geq 6.0$):** 99,998/100,000 random complete codes pass (Stage 2G is nearly vacuous for random codes since hyper-conservative codes are rare in random sampling). The 9 competitors from Phase 5 all pass Stage 2G. Competitors: 9.
- **+Stage 2H (historical reachability ≥ 0.56):** Reduces competitors from 9 to 1. The remaining code has historical reachability 0.571 (just above the 0.56 threshold) but is missing a stop codon.
- **+Stage 2I (all 3 stop codons):** The remaining competitor uses only Stop1 and Stop3, missing Stop2 (UAG). Stage 2I eliminates it. **Competitors remaining: 0.**

Theorem 1 (Computational uniqueness) *Among 10^5 randomly generated wobble-admissible complete genetic codes, no code outperforms the standard genetic code on all eight criteria simultaneously (Stages 2A–2I). The standard code is the unique survivor at this sampling depth.*

The probability of observing 0 competitors by chance (if the standard code were average) is less than 9×10^{-5} , conservatively. The probability that the uniqueness result is an artifact of insufficient sampling decreases as p^{-5} per order-of-magnitude increase in sample size, where $p = 0.009\%$ is the Phase-5 fraction.

4 Discussion

4.1 Two types of “better” codes

Our analysis reveals a qualitative distinction between two types of codes that beat the standard on abstract criteria:

Type I (hyper-conservative): Codes with $\text{max_jump} \leq 5$ found by CP-SAT. These minimize abstract error and clustering scores but prevent the full range of evolutionarily important single-nucleotide substitutions. They represent codes that would be optimal for organisms evolving in hyper-stable environments (e.g., extreme-radiation environments where any mutation is lethal, so only the most conservative code survives). They

are appropriate for *stasis*, not for the evolution of complex life.

Type II (incomplete stop coverage): The one code with 2 stops that passes Stages 2G and 2H. This code is evolutionarily unstable: the missing stop codon (UAG) can be captured by a suppressor tRNA and reassigned to an amino acid, eventually leaving the organism with only one functional stop signal.

Both types reveal scientifically interesting alternatives — the existence of “alien code space” populated by codes optimal for environments radically different from early Earth. Synthetic biology could test whether organisms with Type-I codes show greater robustness to mutation at the cost of reduced evolvability.

4.2 Arithmetic echoes from the GTE canonical orbit

The Universal Generative Principle (UGP) addresses the Standard Model parameter spectrum, classifying each sector by epistemic status [17], via a number-theoretic framework [21, 23]. At the canonical ridge level $n = 10$, the Generative Triple Evolution (GTE) produces an orbit of integer triples (a, b, c) starting from $G_1 = (1, 73, 823)$ via deterministic update rules (see Appendix B). Two arithmetic quantities in this orbit approximate fundamental counts in the universal genetic code:

Echo 1 — Mersenne ratio ≈ 64 codons. The consecutive Mersenne-like c-values $c_2 = 2^{10} - 1 = 1023$ and $c_3 = 2^{16} - 1 = 65535$ satisfy:

$$c_3/c_2 = 65535/1023 = 64 + 63/1023 \approx 64.06 \approx 64 = 4^3.$$

The exact integer division $65535 = 64 \times 1023 + 63$ is Lean 4-certified as `mersenne_ratio_approx` in module `UgpLean.GTE.GeneralTheorems` [24]. The echo is noted in [21] as “codon-count proximity.”

Echo 2 — Shared residue = 20 amino acids. Under the prime-lock condition at $n \geq 10$, the shared residue $m_1 \equiv c_1 \pmod{b_1} = 823 \bmod 73 = 20$ is uniquely forced by the interior divisor condition: any $b_1 > m_1$ gives $m_1 = 20$ exactly. This is Lean-certified as `g1_derived` in module `UgpLean.GTE.UpdateMap` [24]: `gteRemainder 823 73 = 20`. The shared residue $m_1 = 20$ equals the number of standard amino acids.

Echo 3 — $a_2 = 3^2$ and wobble classes = 3^3 . The first derived triple value $a_2 = m_1 - (n + 1) = 20 - 11 = 9 = 3^2$, Lean-certified as `odd_step_a2` [24]. The number of wobble equivalence classes is $27 = 3^3$, with $3 = \sqrt{a_2}$ and also $3 =$ the codon length (triplet). The chain $a_2 = 3^2 \rightarrow 27 = 3^3$ is a structural pattern, not a derivation.

These are *arithmetic echoes*, not derivations. The UGP does not predict the genetic code from first principles. The $m_1 = 20$ echo merits special notice: m_1 is the specific residue that makes $c_1 = b_1 q_1 + 20 = 823$ prime (the prime-lock condition). Among the five residues satisfying all prime-lock constraints at $n = 10$ (namely $\{18, 20, 24, 26, 36\}$), $m_1 = 20$ is uniquely selected by the RSUC-admissible orbit solution $(b_2, q_2) = (42, 24)$. Separately, the Shannon entropy of the polar-requirement distribution of the 20 standard amino acids gives $\exp(H_{\text{PR}}) \approx 19.0$, indicating that the actual code near-maximally utilizes its chemical diversity budget—a self-consistency condition consistent with $A = 20$. Three biological counts (20, 64, 27) from the same GTE orbit as the fine-structure-constant echo $137 \approx 1/\alpha_{\text{EM}}$ [21, 24] constitute a pattern; whether it reflects a common arithmetic self-consistency principle operating at both the particle-physics and molecular-biology scales is an open question.

4.3 Connection to the admissibility-viability convergence pattern

The structural pattern observed here mirrors a result established for the Standard Model of particle physics [22]. In that setting, the SM parameters are the unique arithmetic intersection of a number-theoretic admissibility sieve (GTE triples satisfying discrete constraints) and a physical viability condition (coupling constants, masses, mixing angles matching observation). The present result establishes the same pattern in biology:

1. **Stage 1 (admissibility):** Wobble equivalence reduces $10^{85} \rightarrow 10^{36}$, a 10^{50} -fold reduction. The admissibility condition here is combinatorial (decodability under wobble rules) rather than number-theoretic, but structurally analogous: a large discrete space is collapsed by a consistency constraint.
2. **Stage 2 (viability):** Eight biological criteria applied jointly reduce the feasible set to (approximately) one: the standard code.

The convergence of two independent domains — particle physics and molecular biology — onto this admissibility-viability sieve structure is suggestive of a general principle: *wherever a large discrete space is filtered by the intersection of a structural consistency condition and a multi-criterion viability condition, the surviving set is small or unique.* This generalisation is explored in the UGP programme [21] but the present paper’s results are independent of any specific UGP mechanisms and rest only on the biological criteria described here.

4.4 Limitations

The uniqueness result is computational (sampling-based) rather than analytical. A formal proof of uniqueness

would require either: (a) an exhaustive enumeration of all 5.8×10^{36} admissible codes (computationally infeasible), or (b) an algebraic argument that no complete wobble-admissible code passes all eight criteria while scoring better than the standard. Path (b) would require the Stage 2 criteria to be expressible as polynomial inequalities, allowing application of the Tarski-Seidenberg theorem — an open problem.

The biological justifications for Stages 2G, 2H, 2I are well-motivated but not derived from first principles of prebiotic chemistry. However, a supporting Gen/Drain analysis [19] with published rate constants [15, 12] finds that the minimum enhancement factor for a self-sustaining prebiotic network equals exactly $\text{IPT} \approx 1.1309$: the transition from non-viable to viable chemistry occurs at this specific threshold, consistent with the P15 Information Profit Principle [20]. Rigorous MDL-based versions of the viability criteria from autocatalytic network theory (with experimentally determined rate constants) would complete the formal uniqueness proof and are an open problem.

5 Conclusion

We have shown that the standard genetic code is the unique survivor of a two-stage sieve applied to the full space of wobble-admissible codon tables, under eight simultaneous viability criteria. The five classical criteria (error minimization, prebiotic accessibility, stop robustness, chemical clustering, polar-requirement conservation) place the standard code at $z = +3.22\sigma$ (top 0.009%). Three additional criteria (evolvability, historical reachability, complete stop coverage) eliminate all remaining competitors in a sample of 10^5 complete codes.

The CP-SAT global search reveals that codes mathematically outperforming the standard code all have hyper-conservative mutation profiles ($\max_jump \leq 5$), making them incompatible with the evolution of complex enzymatic diversity. This separates *evolved optimality* (the standard code’s position as the locally-reachable, evolvability-preserving maximum) from *abstract optimality* (the mathematical maximum of simplified criteria).

This result extends to biology a pattern established for the Standard Model [22]: when a large discrete space is filtered by the intersection of a structural admissibility condition and multi-criterion viability, the survivor set is generically small or unique. Whether this reflects a general mathematical principle or a domain-specific coincidence is an open question.

Companion paper. The General Selection paper [18] generalises the two-stage sieve framework used here to eight domains spanning biology, physics, and economics, and situates the genetic-code result within a broader comparative evidence landscape for the General Selection Principle.

6 Robustness and Validation

Null test. We verified that the z-score ordering matches across five independent random seeds ($z = +3.18\sigma$ to $+3.26\sigma$), confirming the result is not seed-dependent. Phase 6 null test: all 9 competitors in Phase 5 were manually examined; their *max_jump* values were in the range [6.5, 8.2] (all pass Stage 2G), and each lacked at least one of: historical reachability ≥ 0.56 or complete stop coverage. No competitor passed all three novel criteria.

Variant code consistency check. All 17 known variant genetic codes (vertebrate mitochondrial, invertebrate mitochondrial, yeast mitochondrial, mold/protozoan, ciliate, Mycoplasma, Spiroplasma, flatworm, etc.) were tested. Every variant scores *below* the standard code on the joint Phase-5 metric. Each variant corresponds to a reassignment of one or two codons; in every case this reassignment is a downhill move on the viability metric, consistent with the variant codes being secondary local minima.

CP-SAT global bound. OR-Tools CP-SAT [14] was run on the 4-criterion and 5-criterion objectives with a 120-second time limit. In both cases, the solver found codes scoring better than the standard code only when *max_jump* ≤ 5 , and the solver status was FEASIBLE (not OPTIMAL). The true global optimum may differ; these are lower bounds on the maximum achievable score among all complete codes. Regardless, all reported CP-SAT codes have *max_jump* < 6.0 and are therefore eliminated by Stage 2G.

Sensitivity to Stage 2G threshold. We scanned $\text{max_jump}_{\min} \in [5.5, 7.0]$. For any threshold ≤ 6.8 , the standard code passes; CP-SAT competitor codes ($\text{max_jump} \approx 4\text{--}5$) are eliminated for any threshold ≥ 5.1 . The result is robust to the precise choice of threshold in this range.

A Numerical Summary Tables

Code Availability

All code is available at:

https://github.com/novaspivack/ugp-physics/papers/25_genetic_code/code/

Key scripts: `codon_sieve.py` (Phase 1 wobble admissibility), `codon_sieve_phase2.py-phase5.py` (Phases 2–5), `codon_sieve_phase6.py` (Phase 6, uniqueness result). Requirements: NumPy, SciPy, OR-Tools (CP-SAT). Runtime: < 5 min on a standard laptop. For details see `papers/25_genetic_code/REPRODUCE.md`.

Table 2: Standard code viability scores on all eight criteria.

Stage	Criterion	Score	Pop.
2A	Error minimization	1.684	top 1%
2B	Prebiotic accessibility	0.697	top 13%
2C	Stop-codon robustness	3 stops	pass
2D	Chemical clustering	0.714	top 20%
2F	Max-jump ≤ 7.40	7.40	boundary
2G	Evolvability ($\text{max_jump} \geq 6$)	7.40	pass
2H	Hist. reachability (≥ 0.56)	0.714	top 15%
2I	All 3 stops present	yes	pass

Table 3: Codes beating the standard as criteria accumulate ($n = 10^5$).

Criteria	Competitors	Fraction
Phase 5 metric	9	0.009%
+ Stage 2G (evolvability)	9	0.009%
+ Stage 2H (reachability)	1	$< 0.002\%$
+ Stage 2I (stop coverage)	0	0%

AI Disclosure

AI assistants (large language models) were used for coding assistance, literature search, and debugging during manuscript preparation. All scientific claims, computational results, and statistical analyses were independently verified by deterministic Python scripts; no AI-generated content appears as unverified scientific output.

Author Contributions

N.S. conceived the framework, designed and implemented all computational analyses (Phases 1–6), performed the CP-SAT global search, wrote the Phase 6 uniqueness code, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

Data Availability

All data generated or analyzed during this study are included in this article and its supplementary code. The amino acid polar-requirement scale (Woese 1965) and prebiotic accessibility rankings used as inputs are publicly available from the cited references.

B The UGP/GTE Framework: A Primer for Biologists

This appendix gives minimal background for readers unfamiliar with the Universal Generative Principle (UGP) and Generative Triple Evolution (GTE), referenced in Section 4.2. The main results of this paper are independent of UGP; the echoes in §4.2 are observations, not derivations. Readers focused on the biological results may skip this appendix.

UGP in one sentence. The Universal Generative Principle [21] is a deterministic, number-theoretic framework proposing that the Standard Model parameter spectrum (fermion masses, gauge coupling constants, mixing angles) can be derived from three axioms: (A1) *Locality*, (A2) *Symmetry*, and (A3) *Compression/MDL* (rules that minimize total description length) [21]. It is complementary to QFT: it fixes the numerical constants that a QFT Lagrangian takes as input; QFT supplies the scattering dynamics. The structural core is machine-checked in Lean 4 (see companion formalization paper [23], zero sorry) [24, 23].

What is GTE? Generative Triple Evolution is the dynamical mechanism realizing the axioms. GTE operates on integer vectors $(a, b, c; g) \in \mathbb{Z}^3 \times \{1, 2, 3\}$: a encodes parity/phase; b is the ladder index (particle’s complexity position and information content $N_{\text{eff}} = |b|$); c is the branch capacity; $g \in \{1, 2, 3\}$ is the Standard Model generation [21]. Two deterministic operators (odd step $g:1 \rightarrow 2$; even step $g:2 \rightarrow 3$) evolve triples with no adjustable parameters.

The prime-lock ridge at $n = 10$. Initial conditions are not free but are uniquely selected by: (i) the ridge $R_n = 2^n - 16$; at $n=10$, $R_{10} = 1008$; (ii) mirror duality: a divisor pair (b_2, q_2) and its swap must both be admissible; and (iii) the prime-lock: $c_1 = b_1 q_1 + 20$ must be prime, where the residue $m_1 = 20$ is uniquely forced by the interior-divisor condition ($b_1 > 20$ at $n \geq 10$) — it is not a free parameter. A systematic search over interior divisors of $R_{10} = 1008$ finds the unique solution $(b_2, q_2) = (42, 24)$, fixing $b_1 = 73$ and the canonical first triple $G_1 = (1, 73, 823; 1)$ (RSUC: Residual Seed Uniqueness Condition, Lean-certified [24, 22]).

The orbit at $n=10$:

$$G_1 = (1, 73, 823; 1) \xrightarrow{\text{odd}} G_2 = (9, 42, 1023; 2) \\ \xrightarrow{\text{even}} G_3 = (5, 275, 65535; 3) \rightarrow \dots$$

This orbit also gives the fine-structure constant echo $2b_1 - a_2 = 2 \times 73 - 9 = 137 \approx 1/\alpha_{\text{EM}}$.

Biological echoes (derived quantities). From $G_1 = (1, 73, 823)$, Lean-certified in `UgpLean.GTE` [24]:

- $q_1 = \lfloor 823/73 \rfloor = 11$; $m_1 = 823 \bmod 73 = 20$ (`g1_derived`; equals **20 amino acids**)
- Odd step: $a_2 = m_1 - (n + 1) = 9$, $b_2 = 73 - 31 = 42$, $c_2 = 2^{10} - 1 = 1023$ (`odd_step_produces_g2`)
- Even step: $c_3 = 2^{16} - 1 = 65535$, $c_3/c_2 \approx 64$ (`mersenne_ratio_approx`; equals **64 codons**)
- $a_2 = 9 = 3^2$; wobble classes $= 27 = 3^3 = 3 \times a_2$

Where to read more. Standard Model derivation: [21]. Uniqueness proof (RSUC): [22]. Lean 4 library (`lake build` to verify): [24]. Formalization paper: [23].

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